# UAV visual localization

Experiment report · ROS 1 Noetic · Individual derivations and team implementation

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The experiment estimates how the camera rotates and moves between consecutive frames. It reuses Lab5 SIFT matches, solves pose with OpenGV, and evaluates it against simulator ground truth.

Data scope: a contiguous prefix of the MIT Office ROS 2 archive, converted to ROS 1. Evaluation covers 213 pairs up to 11.35 s from the first image. Full-sequence evaluation is pending.

| Deliverable | Points | Coverage |
|---|---|---|
| D1 Five-point solver | 10 | Steps, degeneracy, and RANSAC count |
| D2 Minimal solver | 10 | Known-rotation derivation and RANSAC |
| D3 Setup | 10 | ROS, topics, intrinsics, and depth |
| D4 2D–2D | 40 | 5pt, 8pt, 2pt, plus no-RANSAC baseline |
| D5 3D–3D | 30 | Arun + RANSAC and metric error |

Points follow the local course rubric. For team inspection, the demo replays measured RGB, depth and estimated poses.

# D1 · Nistér's five-point solver

Five correspondences constrain a relative pose

Use R and t to map camera 2 into camera 1. For calibrated unit bearings x1 and x2, x1^T E x2 = 0 with E = [t]_cross R. Rotation has three degrees of freedom and translation direction has two.

| Stage | Computation |
| --- | --- |
| 1. Null space | Stack the epipolar constraints into a 5×9 matrix. With rank 5, express E using four null-space basis matrices. |
| 2. Essential constraints | Fix the global scale and impose det(E)=0 and 2EE^TE-tr(EE^T)E=0. Eliminate variables to obtain a degree-ten polynomial. |
| 3. Physical pose | Back-substitute real roots. Decompose E into candidate rotations and ±t, check positive triangulated depth, then score on further correspondences. |

Degeneracy: pure rotation aligns rays after R and leaves translation direction unobservable. Nearly collinear points or very small parallax make estimates unstable. A plane can yield finite solutions, sometimes with multiple compatible candidates. [2]

For independent five-point draws at inlier fraction w, the success probability is about w^5 and the expected wait is 1/w^5. At confidence p, N = ceil(log(1-p) / log(1-w^5)). For w=0.7 and p=0.99, the mean is 5.95 and the budget is 26. The OpenGV NISTER wrapper adds three pairs for decomposition and disambiguation, so it draws eight.

# D2 · Two points with known rotation

Two constraints determine the direction of translation

With R supplied by the IMU, set $n_i = (Rx2_i) \times x1_i$. Each ray pair gives $n_i^T t = 0$, a plane containing translation. The two planes intersect along $t = \pm(n1 \times n2) / ||n1 \times n2||$.

The normals must be independent. Reject zero-parallax or parallel-normal samples. Use cheirality to select the sign. A refinement can take the right singular vector of the inlier-normal matrix with the smallest singular value.

| RANSAC | Procedure |
| --- | --- |
| Sample | Draw two matches, reject degenerate normals, form a direction. |
| Score | Triangulate and use bearing reprojection cost; count residuals below the threshold. |
| Keep | Keep the largest consensus set; use residuals to break ties if needed. Update the budget with $w^2$. |
| Return | Return the best direction and inliers, optionally refit using all inliers. |

The two-point experiment uses ground-truth relative rotation as the known IMU rotation. Its near-zero rotation error follows from this input.

# D3–D5 · Data and implementation

From pixels to relative transforms

| Stage | Implementation and parameters |
| --- | --- |
| Input | Ubuntu 20.04, ROS Noetic, OpenCV 4.2 contrib, OpenGV. Read left RGB, registered depth, CameraInfo and ground-truth odometry. |
| Calibration | fx=fy=415.6922, cx=360, cy=240, zero distortion. Undistort and normalize (x,y,1) to unit bearings. |
| Alignment | Match depth timestamps within 15 ms. Interpolate odometry translation linearly and rotation by SLERP. Apply the starter camera-to-body extrinsic. |
| 2D–2D | Reuse Lab5 SIFT, FLANN and the 0.8 ratio test. OpenGV Nistér, eight-point and known-rotation two-point use RANSAC, threshold 5e-6 in the two-view cost. |
| 3D–3D | Back-project with Z depth and reject invalid samples. Arun aligns centered 3D points by SVD, corrects reflections and recovers metric translation. RANSAC threshold 0.04 m. |
| RPE and RViz | Compare previous-to-current relative transforms. Rotation uses Frobenius distance; 2D translation uses unit directions and 3D uses meters. RViz is anchored to the previous ground-truth pose; 2D display also uses its scale. |

Skip 2D direction error below a 1 mm ground-truth baseline. Failures remain in the CSV as gaps. RANSAC uses up to 300 iterations at 0.99 confidence. The baseline scores one hypothesis from the lowest descriptor-distance matches.

# D4 · 2D correspondence results

Three RANSAC methods and a single-hypothesis baseline

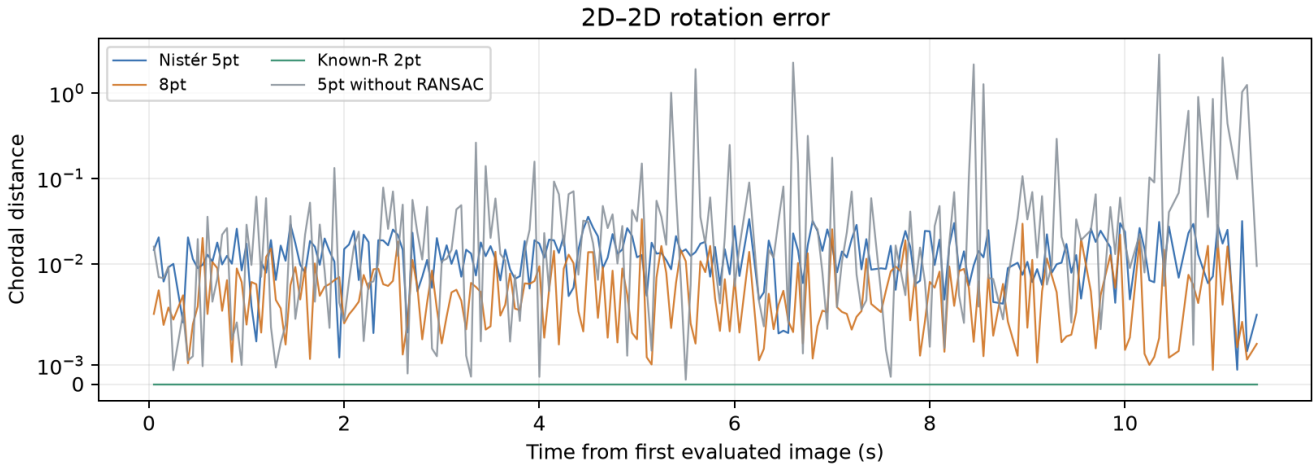



Fig. 05 Chordal rotation error for the 2D methods.

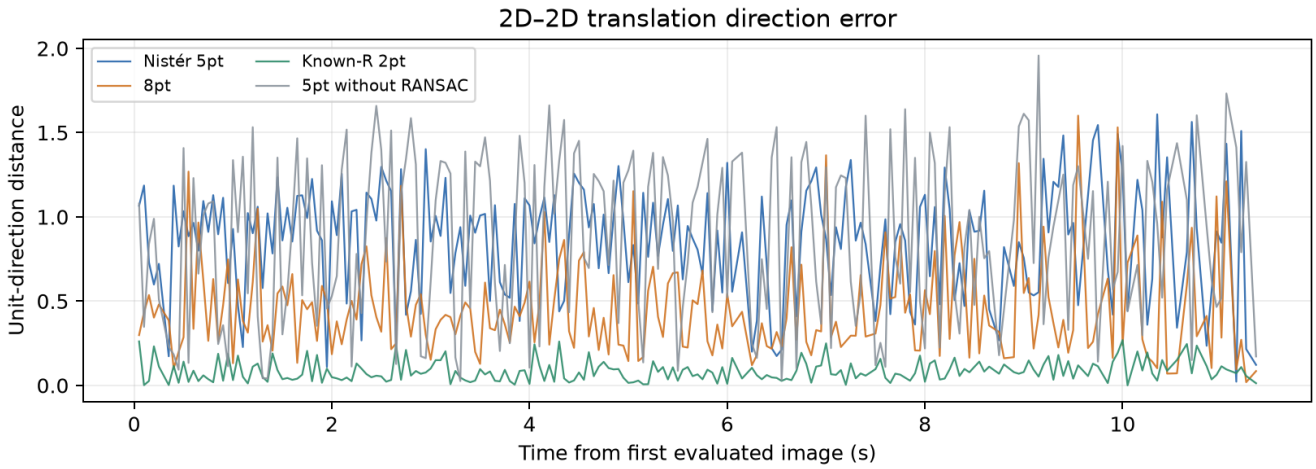



Fig. 06 Translation direction error, ranging from 0 to 2.

| Method | Solved | Dir. samples | Median rot. ° | Median dir. |
|---|---|---|---|---|
| Nistér 5pt | 213/213 | 213 | 0.5296 | 0.9078 |
| 8pt | 213/213 | 213 | 0.1901 | 0.3889 |
| Known-R 2pt | 213/213 | 213 | 0 | 0.07093 |
| 5pt without RANSAC | 213/213 | 213 | 0.6434 | 0.9593 |

With low parallax, small pixel errors can produce large direction errors. Interpret direction and metric errors in their respective units.

# D5 · 3D correspondence results

Arun + RANSAC recovers metric translation

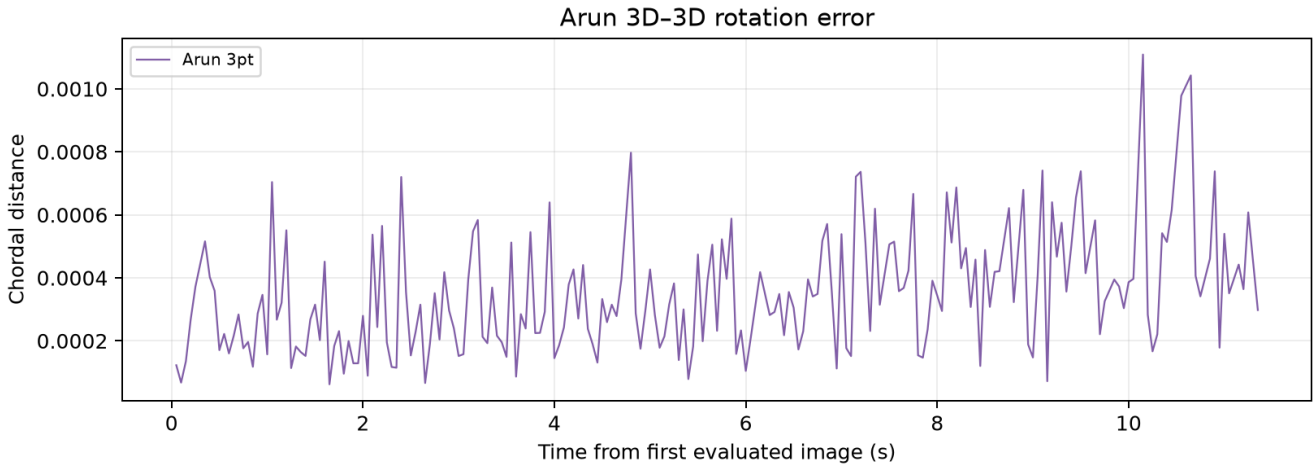



Fig. 07 Arun chordal rotation error.

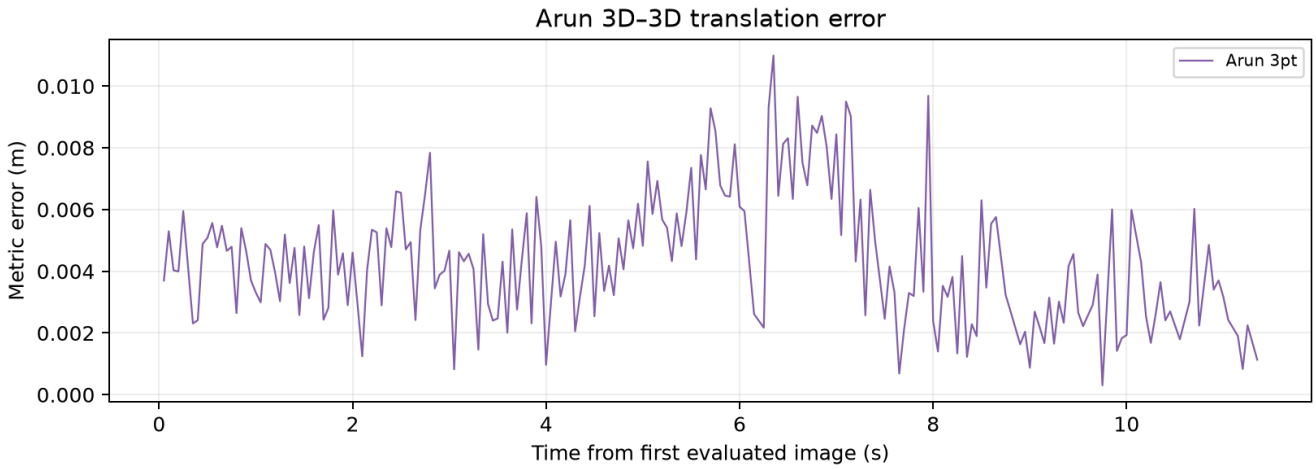



Fig. 08 Arun metric relative translation error.

| Solved | Median rot. ° | Median trans. m | Median inliers | Median solve ms |
|---|---|---|---|---|
| 213/213 | 0.01275 | 0.00429 | 288 | 0.04025 |

Depth turns pixels into points with a metric scale. Metric transforms can be composed across frames, but drift accumulates. The RViz display shows ground-truth-anchored single-step poses.

# RViz replay screenshots

Depth and RGB are on the left; estimated and reference poses are in the center.

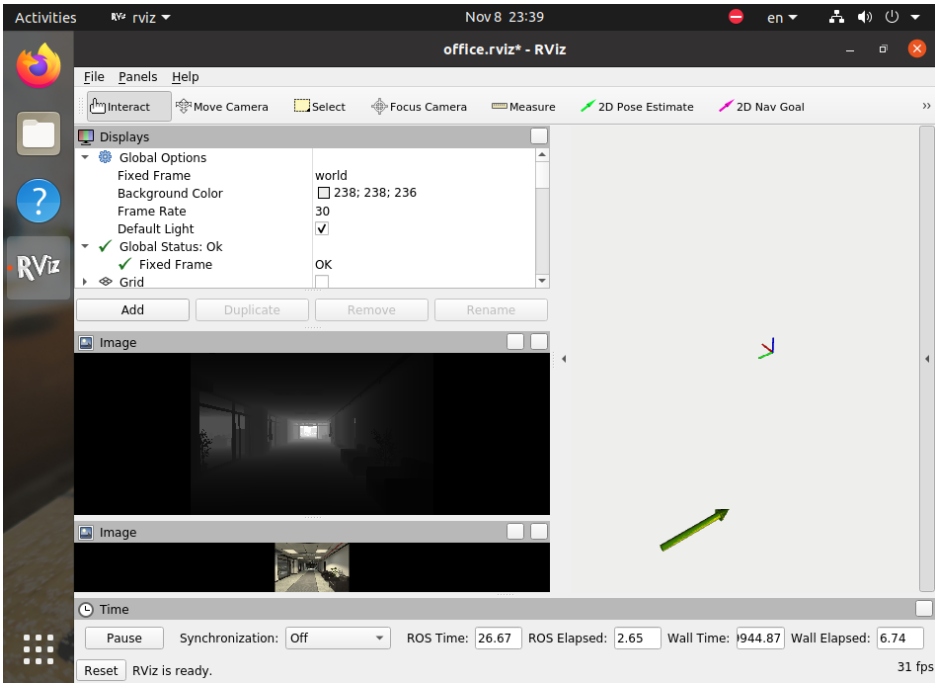



Fig. 01 Nistér 5pt replay of the Office prefix.

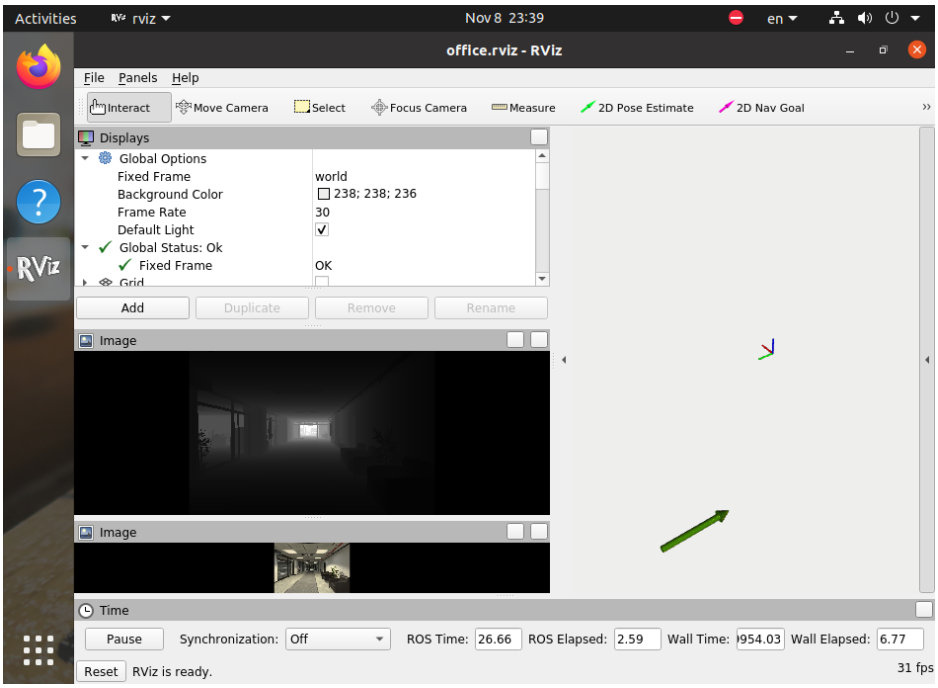



Fig. 02 8pt replay of the Office prefix.

# RViz replay screenshots

Depth and RGB are on the left; estimated and reference poses are in the center.

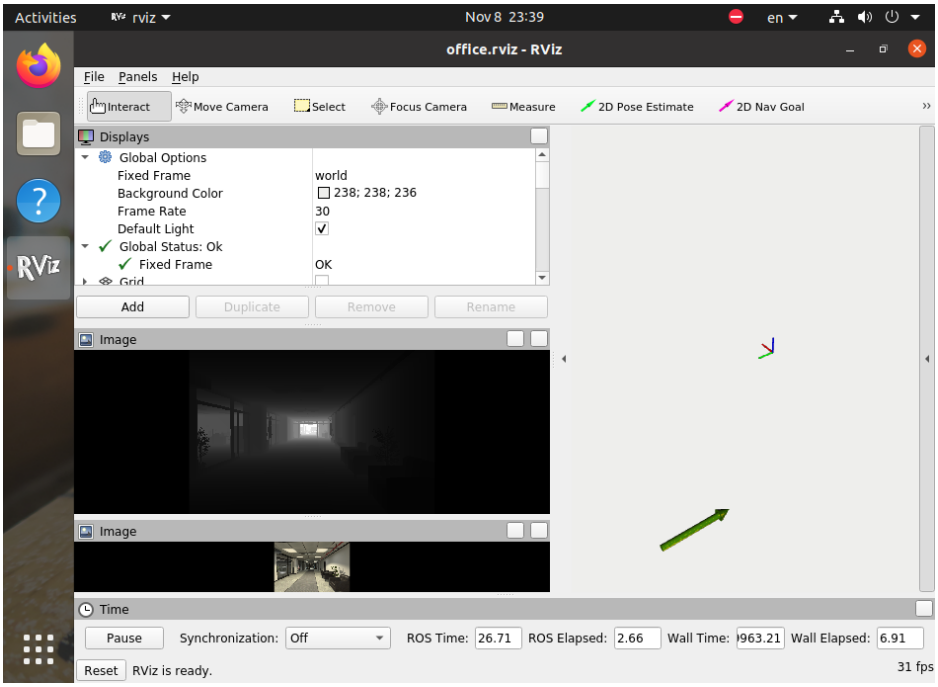



Fig. 03 Known-R 2pt replay of the Office prefix.

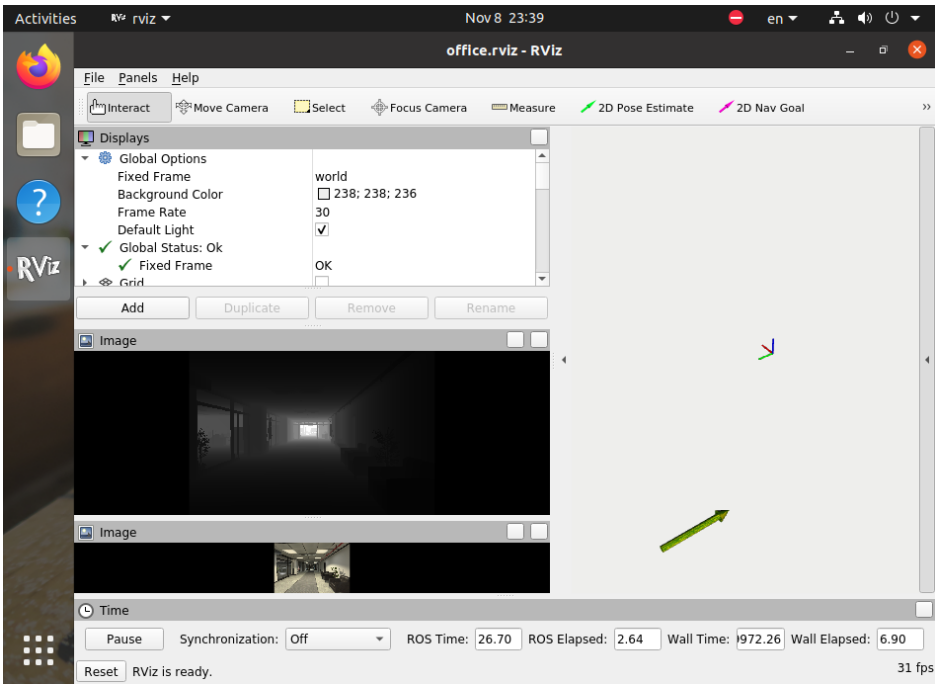



Fig. 04 Arun 3pt replay of the Office prefix.

# Verification and references

Reproduction, metrics and limitations

| Artifact | Purpose |
| --- | --- |
| rpe.csv | Per-pair errors, match counts, inliers, baseline and solver timing. |
| summary.csv | Valid counts and medians. Timing covers pose solving and RANSAC. |
| Demo guide | Run RViz, switch methods, recompute and inspect output. |

Scope: the original Google Drive file returned 404. The current course ROS 2 Office archive supplied the contiguous prefix used here. Full-sequence performance still needs validation. An optional Lab4 RGB-D recording failed during depth camera metadata parsing in the older simulator; its log is retained.

Median 2D direction errors were 0.9078 (5pt), 0.3889 (8pt), 0.07093 (known rotation) and 0.9593 (no RANSAC). Arun's median metric error was 0.00429 m. The translation metrics use different units.

**[1] MIT VNAV, Lab 6 Exercises (ROS 1).**

https://vnav.mit.edu/labs_2023/lab6/exercises.html

**[2] D. Nistér, An Efficient Solution to the Five-Point Relative Pose Problem, CVPR 2003.**

https://www-users.cse.umn.edu/~hspark/CSci5980/nister.pdf

**[3] OpenGV, library and sample-consensus implementation.**

https://github.com/ethz-asl/opengv

**[4] MIT VNAV, Office dataset (ROS 2 archive).**

https://drive.google.com/file/d/1TohOnfjR_FKf_y9wYTEkmv-XCuFPFRPt/view

**[5] MIT-SPARK, VNAV Lab starter code.**

https://github.com/MIT-SPARK/VNAV-labs